Paper Reproduction: SimpleShot Revisiting Nearest-Neighbor Classification for Few-Shot Learning

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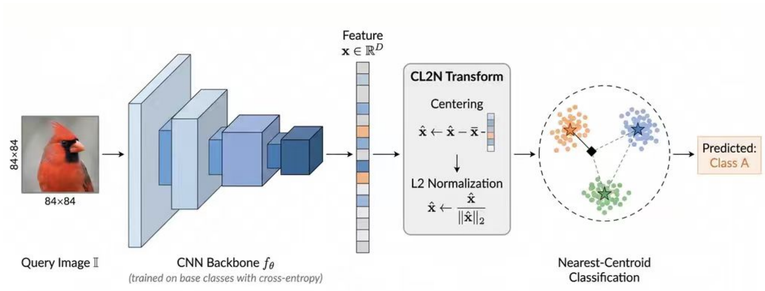
Pipeline



No meta-learning required — simple feature transformations suffice for competitive few-shot learning.

Overall pipeline of the SimpleShot framework from base-class training to nearest-prototype classification.

Questions that we want to explore

1. Can the SimpleShot framework be successfully reproduced on miniImageNet using standard backbone networks?
2. Do deeper networks consistently yield better few-shot classification performance?
3. How much do feature normalization strategies such as L2N and CL2N improve performance over unnormalized features?
4. How different are the final epoch results from the best saved checkpoint results?
5. If our reproduced results fall below the original paper, what likely causes explain the gap?

Experiment Setup

- We conduct all experiments on **miniImageNet**, a standard benchmark for few-shot image classification. The dataset is split into base, validation, and test categories. The backbone is trained on the base split, while few-shot episodes are evaluated on novel classes from the test split.
- We follow the **5-way few-shot** training and testing scheme and for each configuration, **we evaluate the model over 10000 episodes and report the classification accuracy with confidence intervals**.
- The backbone feature extractor models we used are Conv4, ResNet10, ResNet18 [4], and DenseNet121 [5].
- Following the original paper [1], all models were trained for 90 epochs. To better analyze performance differences across models, we evaluated them under two settings. First, we ran the evaluation pipeline on each model after completing all 90 epochs. Second, we repeated the evaluation using the checkpoint corresponding to the epoch that achieved the highest validation accuracy for each model.

Parameters Selection

Following the official SimpleShot configuration [1], we adopted the core hyperparameters including an **SGD optimizer** with momentum 0.9, an **initial learning rate of 0.1** with a MultiStepLR schedule using milestones at [45, 66] and $\gamma = 0.1$, and a **batch size of 256**. All models were trained for 90 epochs using cross-entropy loss on 64 base classes.

Modifications & Refinements: Since the original paper omitted specific details, we implemented these strategies inspired by canonical papers:

- **Standardized Input:** Applied ImageNet normalization ($\mu = [.485, .456, .406], \sigma = [.229, .224, .225]$). This ensures early-epoch loss decreases faster and final features are more discriminative.
- **Data Augmentation:** Used *RandomResizedCrop(84)* and *HorizontalFlip* to learn scale and pose invariance, preventing overfitting on the training split.
- **Weight Decay (1e-4):** Added L2 penalty to SGD to prevent feature geometry elongation. Since SimpleShot relies on Euclidean distance, this improves geometric quality.

Main Results

SimpleShot Replication: All 4 Models vs Paper Table 1 (miniImageNet, 5-way, 10K episodes)

Model	Transform	1-shot Ours	1-shot Paper	Gap	5-shot Ours	5-shot Paper	Gap
conv4	UN	34.28+/-0.17	33.17+/-0.17	+1.11	62.35+/-0.17	63.25+/-0.17	-0.90
conv4	L2N	46.50+/-0.18	48.08+/-0.18	-1.58	64.48+/-0.17	66.49+/-0.17	-2.01
conv4	CL2N	48.03+/-0.19	49.69+/-0.19	-1.66	65.20+/-0.17	66.92+/-0.17	-1.72
resnet10	UN	51.86+/-0.21	54.45+/-0.21	-2.59	74.46+/-0.16	76.98+/-0.15	-2.52
resnet10	L2N	55.68+/-0.20	57.85+/-0.20	-2.17	75.81+/-0.16	78.73+/-0.15	-2.92
resnet10	CL2N	58.39+/-0.20	60.85+/-0.20	-2.46	75.80+/-0.16	78.40+/-0.15	-2.60
resnet18	UN	53.82+/-0.21	56.06+/-0.20	-2.24	76.56+/-0.15	78.63+/-0.15	-2.07
resnet18	L2N	57.72+/-0.20	60.16+/-0.20	-2.44	77.70+/-0.15	79.94+/-0.14	-2.24
resnet18	CL2N	60.30+/-0.20	62.85+/-0.20	-2.55	77.83+/-0.15	80.02+/-0.14	-2.19
densenet121	UN	55.54+/-0.20	57.81+/-0.21	-2.27	77.95+/-0.15	80.43+/-0.15	-2.48
densenet121	L2N	59.55+/-0.20	61.49+/-0.20	-1.94	78.60+/-0.15	81.48+/-0.14	-2.88
densenet121	CL2N	61.78+/-0.20	64.29+/-0.20	-2.51	79.10+/-0.15	81.50+/-0.14	-2.40

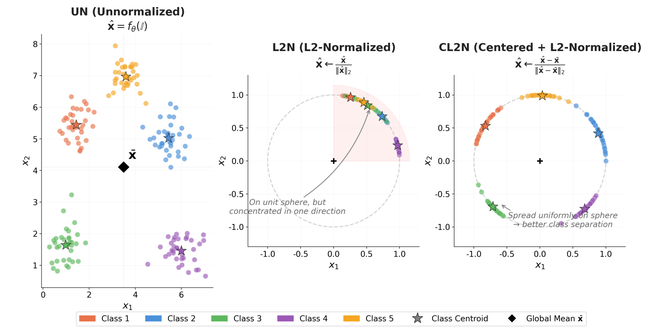
Performance evaluated based on epoch 90 for each model.

SimpleShot Replication: All 4 Models vs Paper Table 1 (miniImageNet, 5-way, 10K episodes)

Model	Transform	1-shot Ours	1-shot Paper	Gap	5-shot Ours	5-shot Paper	Gap
conv4	UN	34.11+/-0.17	33.17+/-0.17	+0.94	62.19+/-0.17	63.25+/-0.17	-1.06
conv4	L2N	46.34+/-0.18	48.08+/-0.18	-1.74	64.52+/-0.17	66.49+/-0.17	-1.97
conv4	CL2N	47.98+/-0.19	49.69+/-0.19	-1.71	65.12+/-0.17	66.92+/-0.17	-1.80
resnet10	UN	52.01+/-0.21	54.45+/-0.21	-2.44	74.39+/-0.16	76.98+/-0.15	-2.59
resnet10	L2N	55.76+/-0.20	57.85+/-0.20	-2.09	75.77+/-0.15	78.73+/-0.15	-2.96
resnet10	CL2N	58.33+/-0.20	60.85+/-0.20	-2.52	75.75+/-0.16	78.40+/-0.15	-2.65
resnet18	UN	53.90+/-0.21	56.06+/-0.20	-2.16	76.47+/-0.15	78.63+/-0.15	-2.16
resnet18	L2N	58.05+/-0.20	60.16+/-0.20	-2.11	77.59+/-0.15	79.94+/-0.14	-2.35
resnet18	CL2N	60.34+/-0.20	62.85+/-0.20	-2.51	77.61+/-0.15	80.02+/-0.14	-2.41
densenet121	UN	55.40+/-0.20	57.81+/-0.21	-2.41	78.17+/-0.15	80.43+/-0.15	-2.26
densenet121	L2N	59.64+/-0.20	61.49+/-0.20	-1.85	78.81+/-0.14	81.48+/-0.14	-2.67
densenet121	CL2N	61.94+/-0.20	64.29+/-0.20	-2.35	79.08+/-0.15	81.50+/-0.14	-2.42

Performance evaluated based on the best epoch for each model.

UN L2N CL2N Visualization



Visualization of how UN, L2N, and CL2N transform feature geometry and improve class separation for nearest-prototype classification.

Observation

- Our reproduction reveals **several consistent patterns** across all backbone architectures and evaluation settings.
- **Performance improves steadily with model depth**. Conv4 performs the worst, while DenseNet121 achieves the best results. This suggests that stronger feature representations directly improve few-shot generalization.
- **Feature normalization plays a crucial role**. Across all models, L2N consistently outperforms unnormalized features, while CL2N further improves performance in most cases.
- **The final checkpoint and the best saved checkpoint show only small differences**. This suggests that the training process is stable and does not show significant overfitting near the end of training.
- Although our results reproduce the overall trends reported in the original paper, the absolute performance is lower by about **2–3%** for most settings. This gap is likely due to implementation differences, including the PyTorch version and dataset preprocessing pipeline.

References

1. Y. Wang, W.-L. Chao, K. Q. Weinberger, and L. van der Maaten, "SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning," *arXiv preprint arXiv:1911.04623*, 2019.
2. O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching Networks for One Shot Learning," in *NeurIPS*, 2016.
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4. K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *CVPR*, 2016.
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